Muhammad Ghulam Mujtaba 24K-0535

Date:

OS- lab 7

Task 1

Process	AT	BT	CT	TAT	WT
P_1	0	6	6	6	0
P_2	2	8	14	12	4
P_3	4	7	21	17	10
P_4	6	3	24	18	15

Gant chart

P_1	P_2	P_2	P_4

0 6 14 21 24

$$TAT = \frac{\Sigma T}{n} = \frac{6+12+17+18}{4} = 13.25 \text{ unit}$$

$$\text{avg } WT = \frac{0+4+10+5}{4} = 7.25 \text{ unit}$$

Task 2

SJF NON- Preemptive

Process	AT	BT	CT	TAT	WT
P_1	1	5	6	5	0
P_2	3	2	8	5	3
P_3	5	9	21	16	7
P_4	6	4	12	6	2

Date:

Grant chart

idle	P_1	P_2	P_4	P_3

$t=0$ 1 6 8 12 21

$$\text{avg TAT} = \frac{5+5+16+6}{4} = 8 \text{ unit}$$

$$\text{avg WT} = \frac{0+3+7+2}{4} = 3 \text{ unit}$$

Task 3

Grant chart

P_1	P_2	P_2	P_2	P_2	P_4	P_4	P_3	P_1

$t=0$ 1 2 3 4 5 6 7 14 22

$$\text{avg TAT} = \frac{22+4+12+4}{4} = 10.5 \text{ unit}$$

$$\text{avg WT} = \frac{13+0+5+2}{4} = 5 \text{ unit}$$

Task 4

Process	AT	BT	CT	TAT	WT
P_1	0	10	26	26	16
P_2	1	5	17	16	11
P_3	2	8	25	23	15
P_4	3	3	12	9	6

Date:

Grant chart

P1	P2	P3	P4	P1	P2	P3	P1	P3	P1

t=0 3 6 9 12 15 17 20 23 26 26

$$\text{avg TAT} = \frac{26+16+23+19}{4} = 10.5 \text{ unit}$$

Task 5

Process	AT	BT	CT	TAT	WT
P1	0	12	36	36	18
P2	2	6	18	16	10
P3	4	8	28	24	16
P4	6	4	20	14	10

Grant chart

P1	P2	P1	P3	P2	P4	P1	P3	P2

t=0 2 4 6 8 10 12 14 16 18

P4	P1	P3	P1	P3	P1

20 22 24 26 28 30

$$\text{avg TAT} = \frac{30+16+24+14}{4} = 21 \text{ unit}$$

$$\text{avg WT} = \frac{18+10+16+10}{4} = 13.5 \text{ unit}$$

Date:

Task 6

Process	AT	BT	CT	TAT	WT
P_1	0	7	7	7	0
P_2	1	4	11	10	6
P_3	2	6	17	15	9
P_4	3	5	22	19	14

Grant chart

P1	P2	P3	P4

t=0 7 11 17 22

$$\text{Avg TAT} = \frac{7+10+15+19}{4} = 12.75 \text{ unit}$$

$$\text{Avg WT} = 0+6+9+14 = 7.25 \text{ unit}$$

Task 7

Process	AT	BT	CT	TAT	WT
P1	0	8	11	11	3
P2	1	3	4	3	0
P3	2	5	16	14	9

grant chart

P1	P2	P1	P3

t=0 1 4 11 16

$$\text{Avg TAT} = \frac{11+3+14}{3} = 9.33 \text{ unit}$$

$$\text{Avg WT} = \frac{3+0+9}{3} = 4 \text{ unit}$$